东 南 大 学 考 试 卷（ A 卷）

课程名称 离散数学 考试学期 2020-2021-3 得分

适用专业 计算机类 考试形式 闭卷 考试时间长度 120 分钟

（开卷、半开卷请在此写明考试可带哪些资料）

题目	1	2	3	4	5	6	7	8	9	10	总分
得分											
批阅人											

自觉遵守考场纪律 如考试作弊 此答卷无效

姓名_____ 学号_____

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1. Choice questions （ 2 points for each question ）

(1) Which of the following compound propositions is a tautology? ( )

A. $(p \wedge q) \rightarrow (p \vee q)$

B. $p \wedge (p \vee q) \leftrightarrow \neg p$

C. $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$

D. $(p \wedge q) \rightarrow \neg p$

(2). What is the false equation about sets? ( )

A. $A-B=A\cap\overline{B}$.

B. $A \cup B = \{x|x \in A \vee x \in B\}$.

C. $A \cap (B \cup C) = (A \cap B) \cup C$.

D. $|A \cup B| = |A| + |B| - |A \cap B|$.

(3). The set A={1, 2, 3}，and we have the relation R={<1,1>, <2,2>, <2,3>, <3,2>, <3,3>}, then R is ( ).

A. Reflexive. B. Symmetric.

C. Transitive. D. All of the above.

(4). Suppose there is a binary operation "*" defined over positive real number set R^+ as $x*y=\frac{x+y}{xy}$. Then, the operation satisfies which one of the following properties? ( )

A. Commutative Law B. Idempotent Law

C. There exists an identity element D. There exists zero element

(5). Which one of the following statements is true? ( )

A. The set of integers "Z" with the minus operation "-" is a semi group <Z, ->.

B. The set of integers "Z" with the minimum operator "min" is a monoid.

C. The set of non-negative integers "$Z^+ \cup \{0\}$" with the maximum operator "max" is a monoid.

D. The set of natural number "N" with the addition operation "+" is a group.

(6). Suppose G is a connected planar simple graph with 5 vertices and 6 regions (i.e. surfaces), then G has _____ edges. ( )

A. 7 B. 8 C.9 D. 10

2. Completion （2 points for each question）

(1). Let *C(x)*, *P(x)*, *D(x)*, and *S(x, y)* be the statements "x is a student in the class," "x is a programming language," "x has studied discrete mathematics," "x has studied y," respectively. Using quantifiers to express each of these statements.

a) Every student in the class has studied discrete mathematics.

_____.

b) Some students in the class have studied all the programming languages.

_____.

(2). With *n* propositional variables, we can construct_____distinct (that is, not equivalent) propositions.

(3). The set A has n elements, then the number of the proper subsets of A is _____.

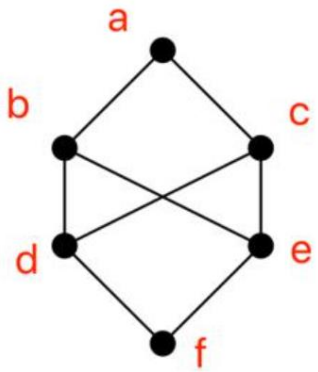
(4). The set A={1, 2, 3, 4, 5}， and R is a binary relation of A, R={<1, 1>, <2, 3>, <3, 4>, <3,2>, <1,5>, <5,5>}， then the reflexive closure of the relation R = _____

_____, and the symmetric closure of the relation R = _____,

and the transitive closure of the relation R = _____

_____.

(5). Given the poset denoted by the following Hasse diagram, the upper bound of elements {b, d, e} is _____.



(6). A simple graph G with 9 vertices is connected if it has more than _____ edges.

3. Find the Disjunctive Normal Form of the following formula. (6 points)

$((\neg p \rightarrow q) \wedge r) \rightarrow \neg q$

4. Construct a valid argument with premises $(p \wedge t) \rightarrow (r \vee s)$, $q \rightarrow (u \wedge t)$, $u \rightarrow p$, $\neg s$ and conclusion $q \rightarrow r$. (9 points)

5. Let *m* be an integer with $m > 1$. Show that the relation $R = \{(a,b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers. （9 points）

6. The set S={1, 2, 5, 7, 10, 14, 35, 70}, the binary relation R is defined as the divisibility relation (|) on S.

(1) Show that R is a partial ordering on the set of S and draw the Hasse Diagram.（5 points）

(2) If T = S – {1, 70}, which elements of the poset <T, | > are maximal, and which are minimal?（4 points）

7. Given a binary operation "*" defined as the following table, answer the questions.

*	a	b	c	d
a	a	a	a	a
b	a	(1)	c	d
c	a	c	c	b
d	(2)	d	b	d

(1) If the operation satisfies the idempotent law, which element should be filled in blank (1)? （2 points）

(2) If the operation satisfies the commutative law, which element should be filled in blank (2)? （2 points）

(3) Continue after question (1) and (2), is there the identity element or zero element? If exists, please identify. （2 points）

(4) Continue after question (1) and (2), is the operation associative? Please explain. （3 points）

8. If group <G, *> has two sub-groups <H, *> and <K, *>, answer the following questions and justify your answers.

(1) Does <H∩K, *> must be a sub-group of group G? （3 points）

(2) Does <H∪K, *> must be a sub-group of group G? （3 points）

(3) If |H|=p, |K|=q, and gcd(p, q)=1 (i.e. p and q are co-prime), prove |H∩K|=1. （4 points）

9. Let L = { x | x∈Z ∧ |x| ≤ 100}, where ≤ is the less than or equal relation on the set of integers Z. For the totally ordered set < L, ≤ >, prove < L, ≤ > is a lattice, and determine whether < L, ≤ > is a distributive lattice or a complemented lattice. Please justify your answers. （9 points）

10. For $G = K_{m,n}$,

(1) Determine when G has an Euler circuit and when G has a Hamilton circuit, justify your answer. (4 points)

(2) If m = 3 and n = 3, prove G is not a planar graph. (5 points)